



University of Idaho

Department of Computer Science

CS 4727/5727
Adversarial
Machine Learning

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Lecture 10

AML in Cybersecurity – Part III: Spam Filtering, URL Detection, Cyber- Physical Systems



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Spam Filtering

10.4 Spam Filtering

- The purpose of a *spam filter* is to determine whether an incoming message is legitimate (i.e., non-spam, **ham**) or unsolicited (i.e., **spam**)
- ML-based spam classifiers were among the first applications of machine learning in the cyber security domain (and in general, as well)
 - Subsequently, they were among the first to be attacked
 - Attackers' goal is to modify spam emails (without changing the nature of the message) to bypass spam filters
- Recent spam filters increasingly rely on machine learning and neural networks approaches for email classification
 - These approaches are being extensively used by email service providers like Gmail, Outlook, or Yahoo

Spam Filtering Datasets

10.4.1 Malware Spam Filtering Datasets

- Examples of open datasets of spam and non-spam email messages
- *TREC 2007 Public Spam Corpus* ([link](#))
 - Also known as Trect07p, created in 2007
 - Consists of 50,200 spam emails (67%) and 25,200 ham (non-spam) emails (33%)
 - The format of the emails is raw data (HTML)
- *Enron-Spam dataset* ([link](#))
 - Created in 2006
 - Includes 20,170 spam emails (55%) and 16,545 non-spam emails (45%)
 - Both raw messages and pre-processed messages are provided
- *SPAMBASE* ([link](#))
 - Created in 1999
 - Includes 1,813 spam emails (39%) and 2,788 non-spam emails (61%)
 - Each email has 55 features, related to word frequency, character frequency, average length of uninterrupted sequence of capital letters, total number of capital letters, etc.



Spam Filtering Techniques

10.4.2 Spam Filtering Techniques

- Based on the used email filtering techniques, spam detectors can be generally classified into:
 - Content-based filtering techniques
 - Heuristic-based filtering techniques

Content-based Filtering

10.4.2 Spam Filtering Techniques

- *Content-based filtering techniques*
 - The filter scans the content of incoming emails, looking for trigger keywords
 - E.g., keywords frequently used in spam emails, such as *free*, *buy*, *application*, *mortgage*
 - The content of the body and header of emails are scanned
- The frequency of occurrence and distribution of trigger words and phrases in the content of emails are used as features for training ML approaches, and afterward, for classifying new emails
 - Naïve Bayes classifiers were one of the early successful ML models for spam filtering
 - Other conventional ML approaches have been successfully applied, such as SVMs, *k*-nearest neighbors, decision trees, random forests
 - NNs and deep learning are commonly used nowadays for spam classification
- Almost all commercial spam filters use some form of content-based filtering
 - A limitation of this approach is that harmless emails containing spam trigger words can be blocked

Content-based Filtering

10.4.2 Spam Filtering Techniques

- Scanning the body of emails explores the **what** in the email
 - Scanning the header of emails explores the **who** sent the email
- Email headers display important information, such as:
 - Message ID – an identifier generated by the sender’s email service
 - There can be no two identical message IDs, hence, it helps to detect forged email headers
 - Sender address – is used to consult black lists to check sender’s domain reputation
 - DNS records – the DNS (Domain Name System) records of the sender allows to check the sender’s SPF, DKIM, and DMARC policies regarding email authentication
 - SPF (Sender Policy Framework), DKIM (Domain Keys Identified Mail), DMARC (Domain-based Message Authentication, Reporting and Conformance)
- An example of a Gmail header

Original Message

Message ID	<000000000000d7d44705c2854da2@google.com>
Created at:	Mon, May 17, 2021 at 2:57 PM (Delivered after 12 seconds)
From:	christian@gmail.com
To:	folderly@gmail.com
Subject:	Request to connect
SPF:	PASS with IP 209.85.220.69 Learn more
DKIM:	'PASS' with domain belkins.io Learn more
DMARC:	'PASS' Learn more

Heuristic-based Filtering

10.4.2 Spam Filtering Techniques

- *Heuristic-based filtering techniques*
 - These approaches apply heuristic rules to discover similar patterns in a large number of spam and non-spam emails
 - Scores are assigned to each rule, and the scores are weighted based on the importance of the rule
 - Repeating patterns in a message increase the total score of being a spam
 - If the total score surpasses a predefined threshold, the message is labeled as spam
- Rules could be created based on:
 - Words and phrases, lots of uppercase characters, exclamation points, unusual Subject lines, special characters, web links, HTML messages, background colors, etc.
- Content-based filtering techniques can be considered a sub-category of the heuristic-based techniques
- A limitation of this approach is that it requires constant updating of the rules, to be able to cope with the continually adapting strategies by spammers



Black Lists and White Lists

10.4.2 Spam Filtering Techniques

- **Black lists** contain information of known spammers, collected by several sites
 - Senders of incoming emails are compared to the blacklist, to filter known spammers
- **White lists** are complementary to black lists, and contain addresses of trusted contacts
- Black lists and white lists are used for the first level of spam filtering
 - E.g., before applying content checks or heuristic rules
 - That is, the lists are used as a complementary tool, and not as the only tool for email classification

Feature Extraction

10.4.2 Spam Filtering Techniques

- Traditional ML-based spam filters extract a custom set of features for training an ML model
- The extracted features from emails can include:
 - **Body-based features**: features extracted from the email message content
 - **Subject line-based features**: features extracted from the subject line of the email
 - **Sender address-based features**: features extracted from the information about the email address of the sender
 - **URL-based features**: features extracted from the anchor tags of HTML emails
 - **Script-based features**: features extracted from the information concerning the presence or absence of scripts in the email and the impact of such scripts

Feature Extraction

10.4.2 Spam Filtering Techniques

- Example: 40 features extracted from emails, categorized based on the information from the previous page
 - The features can be used to train an ML model for classification of spam messages

Feature category	Feature	Feature type	Summary
Body	html	Binary	Presence or absence of HTML tags in the body
	forms	Binary	Presence or absence of forms in the body
	numWords	Continuous	Total number of words in the body
	numCharacters	Continuous	Total number of characters in the body
	numDistinctWords	Continuous	Total number of distinct words in the body
	richness	Continuous	Ratio of numWords to numCharacters in the body
	numFunctionWords	Continuous	Total occurrence of keywords such as account, access, bank, click, credit, identity, information, inconvenience, limited, log, minutes, password, risk, recently, social, security, service, and suspended in the body
	suspension	Binary	Presence or absence of the word 'suspension' in the body
	verifyYourAccount	Binary	Presence or absence of the phrase 'verify your account'
Subject line	reply	Binary	Checks if the email is a reply to a previous mail
	forward	Binary	Checks if the email is forwarded from another account
	numWords	Continuous	Total number of words in the subject line
	numCharacters	Continuous	Total number of characters in the subject line
	richness	Continuous	Ratio of numWords to numCharacters in the subject line
	verify	Binary	Presence or absence of the word 'verify' in the subject line
	debit	Binary	Presence or absence of the word 'debit' in the subject line
	bank	Binary	Presence or absence of the word 'bank' in the subject line

Feature category	Feature	Feature type	Summary
Sender address	numWords	Continuous	Total number of words in the sender address field
	numCharacters	Continuous	Total number of characters in the sender address field
	diffSenderReplyTo	Binary	Checks if the sender's domain and reply-to domain are different
	nonModalSenderDomain	Binary	Checks if the sender's domain and email's modal are the same
URL	ipAddress	Binary	Checks for the use of IP address rather than a qualified domain
	numIpAddresses	Continuous	Number of links with IP addresses and not domain names
	atSymbol	Binary	Presence of links that contain an '@' symbol.
	numLinks	Continuous	Total number of links in the email body
	numInternalLinks	Continuous	Total number of links in the body with internal targets
	numExternalLinks	Continuous	Total number of links in the body with external targets
	numImageLinks	Continuous	Total number of links in the body with an image
	numDomains	Continuous	Total number of domains from all the URLs in the body
	maxNumPeriods	Continuous	Highest number of periods from all the links
	linkText	Binary	Checks if the link text contains words like click, here, login, or update
	nonModalHereLinks	Binary	Checks for 'here' links mapping to a non-modal domain
	ports	Binary	Checks for URLs accessing the ports other than 80
Script	numPorts	Continuous	Number of links in the email with the port information
	scripts	Binary	Presence or absence of scripts in the body
	javaScript	Binary	Presence or absence of JavaScript in the body
	statusChange	Binary	Checks if any script overwrites the status bar of the email client
	popups	Binary	Presence or absence of any popup code in the body
	numOnClickEvents	Continuous	Total number of onClick events in the body
	nonModalJsLoads	Binary	Checks for any non-modal external JavaScript forms

Spam Filtering with Naive Bayes

10.4.2 Spam Filtering Techniques

- *Naive Bayes classifier* has been one of the most popular ML models for spam filtering
 - It is easy to implement, has low computational complexity, and provides statistical measure of the probability that a message is spam or non-spam
- From Bayes' theorem, the probability that an email message represented with a vector $\mathbf{x} = [x_1, x_2, \dots]$ belongs to the spam category c_s is

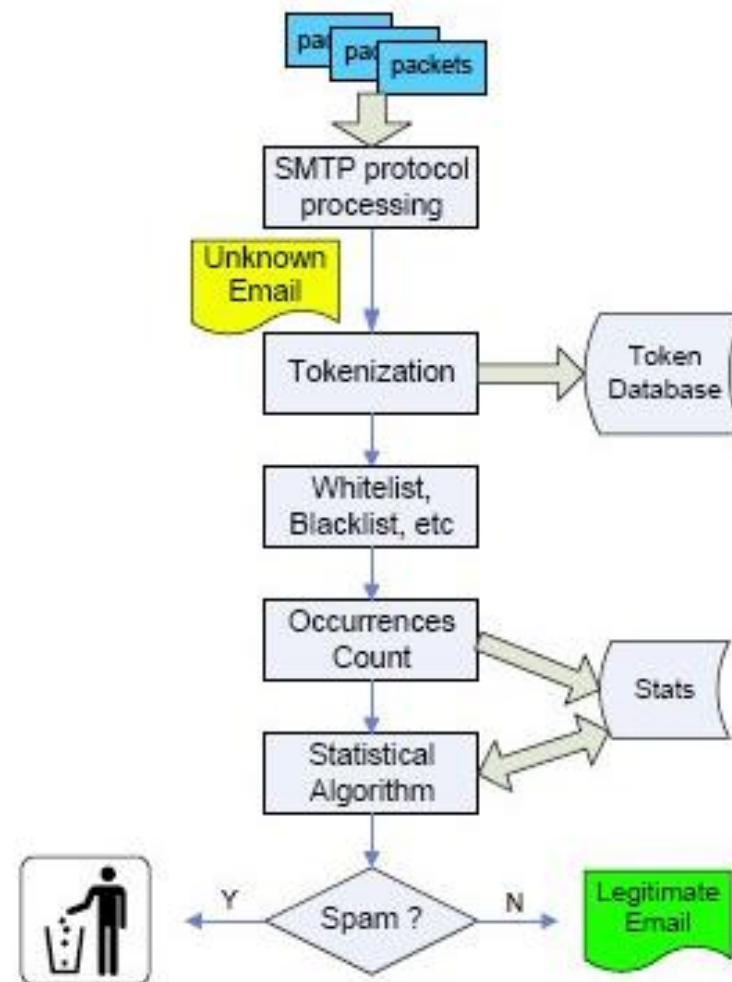
$$p(c_s|\mathbf{x}) = \frac{p(c_s) \cdot p(\mathbf{x}|c_s)}{p(c_s) \cdot p(\mathbf{x}|c_s) + p(c_h) \cdot p(\mathbf{x}|c_h)}$$

- In the equation c_h is the ham category
- The prior probabilities $p(c_s)$ and $p(c_h)$ are typically estimated by dividing the number of training emails in each category by the total number of training emails
- The probabilities $p(\mathbf{x}|c_s)$ and $p(\mathbf{x}|c_h)$ are calculated as a product of the probability that each feature belongs to the spam or ham bag-of-words, i.e., $p(\mathbf{x}|c_s) = \prod p(x_i|c_s)$
- If $p(c_s|\mathbf{x}) > \text{threshold}$, the email message is classified as a spam

Spam Filtering Block Diagram

10.4.2 Spam Filtering Techniques

- A typical data flow in spam filtering





Spam Filtering with Sequence Models

10.4.2 Spam Filtering Techniques

- Sequence models process the text in message directly for classification of spam and non-spam messages
 - Commonly used sequence models are Recurrent Neural Networks and Transformer Networks
- The application of sequence models typically involves:
 1. Tokenization to represent the words in text data with integer indices
 2. Mapping the integers to vector representations (embeddings)
 3. Pad the sequences in the text to have the same length
 4. Use the padded sequences as inputs to train a machine learning model

Commercial Spam Filters

10.4.2 Spam Filtering Techniques

- Examples of three different spam filters solutions
 - Gmail and Outlook use these spam filters

Barracuda	An on-premise, cloud-based spam filter that has a constantly growing database of notorious spammers. Barracuda refers to this database to make sure that incoming messages don't belong to spam senders or malware websites. Additionally, it locates spam texts hidden inside images.
Microsoft Outlook	A built-in appliance responsible for protecting the email client from spam attacks. It checks the content of incoming emails, scans senders' DNS records, and redirects suspicious emails to a spam folder.
Spam Assassin	A standalone program that can be integrated with your mail server. Spam Assassin uses a wide set of techniques, from checking DNS records to Bayesian filtering, in order to filter all incoming messages. Can be used by one user or a business network.

AML against Spam Filters

10.4.3 Adversarial Attacks Against ML-based Spam Filters

- Common approaches for creating adversarial attacks against ML-based spam filters include:
 - *Bad words obfuscation* – replace typical words in spam messages with synonyms or misspelled words
 - *Good words insertion* – insert into spam messages words that appear in legitimate messages
- *Huang et al. (2011) Adversarial Machine Learning* ([link](#))
 - This work introduced an availability attack
 - I.e., the attack makes the spam filter unavailable for regular use
 - Attacked is a model called SpamBayes, which uses Naïve Bayes for spam filtering based on words occurrences in the content of email messages
 - Attack approach:
 - Send spam email messages that contain a very large set of words to the spam filter
 - The spam filter algorithm will recognize the emails as spam, and it will assign a higher spam score to every word in the received messages
 - As a result, future legitimate emails are more likely to be marked as spam

AML against Spam Filters

10.4.3 Adversarial Attacks Against ML-based Spam Filters

- *Biggio et al. (2014) Security Evaluation of Pattern Classifiers under Attack* ([link](#))
 - White-box availability attack
 - Attacked are two spam classifiers: linear SVM and logistic regression classifier
 - The classifiers used bag-of-words representation based on text content, where features are word occurrences
 - Attack approach:
 - Use an optimization approach to find most impactful non-spam words
 - Add $n_{\max}$ most impactful non-spam words to spam emails
 - This can cause the spam filter to increase the scores for the impactful good words, and as a result, the model will be more likely to classify non-spam emails as spam

AML against Spam Filters

10.4.3 Adversarial Attacks Against ML-based Spam Filters

- *Sethi and Kantradzic (2018) Data Driven Exploratory Attacks on Black Box Classifiers in Adversarial Domains* ([link](#))
 - Black-box query-efficient evasion attacks
 - The authors trained 5 different spam classifiers using conventional ML methods: linear SVM, k -nearest neighbors, SVM with RBF kernel, decision tree, and random forest
 - SPAMBASE dataset was used for model training and evaluation
 - The authors introduced a framework called **SEE (Seed-Explore-Exploit)**
 - Attacks include:
 - Anchor Points (AP) attack: anchors are legitimate emails, that serve as a ground-truth for generating adversarial samples by applying perturbation to the legitimate emails
 - The spam filter is queried, and the procedure is repeated until the emails are classified as spam
 - Reverse Engineering (RE) attack: the goal is to discover the decision boundary for spam email classification, and ultimately learn a substitute model based on querying the target classifier
 - The generated samples against the substitute model are then transferred to the target classifier

AML against Spam Filters

10.4.3 Adversarial Attacks Against ML-based Spam Filters

- *Cheng et al. (2022) Adversarial Email Generation against Spam Detection Models through Feature Perturbation* ([link](#))
 - White-box evasion attack against ML-based spam email classifiers
 - Datasets: Enron email corpus and TREC 2007 Public Spam Corpus
 - Attack approach:
 - Compute adversarial perturbations in feature space using bag-of-words representation
 - Translate perturbations back to legitimate-looking words inserted into the spam email body
 - These words change the model confidence from spam to non-spam without altering the main message intent
 - Evaluated against SVM, Random Forest, and Logistic Regression classifiers
 - Achieved substantial decrease in spam detection accuracy for all evaluated ML-based spam email classifiers

AML against Spam Filters

10.4.3 Adversarial Attacks Against ML-based Spam Filters

- *Roumeliotis et al. (2024) Next-Generation Spam Filtering: Comparative Fine-Tuning of LLMs, NLPs, and CNN Models for Email Spam Classification* ([link](#))
 - Compares finetuned GPT-4, BERT, RoBERTa and CNN for email spam classification
 - Approach:
 - Finetune LLMs and NLP models on labeled spam and non-spam datasets using few-shot learning
 - Two datasets: Dataset 1 (3,665 emails) and Dataset 2 (3,565 emails)
 - Pre-finetuning and post-finetuning performance compared for all models
 - CNN model serves as the traditional baseline for comparison
 - Post-fine-tuning accuracy results:
 - GPT-4: 99.30% (Dataset 1), 98.29% (Dataset 2)
 - BERT: 99.39% (Dataset 1), 99.01% (Dataset 2)
 - RoBERTa: 99.04% (Dataset 1), 98.38% (Dataset 2)
 - CNN (baseline): 76.09% (Dataset 1), 86.62% (Dataset 2)
 - Transformer-based models demonstrate better robustness to spam email than traditional classifiers based on CNN models

URL Detection

10.5 URL Detection

- **URL (Uniform Resources Locator)** is a web address that specifies the location of the webpage on a computer network
- A typical URL <http://www.example.com/index.html> consists of several components:
 - **Protocol type** = http
 - **Domain name** = www.example.com
 - **File name** = index.html
- The domain name is also referred to as **FQDN (Fully Qualified Domain Name)**
 - It identifies the server hosting the webpage
 - FQDN can be further divided into a **prefix (subdomain)** = www, and an **RDN (Registered Domain Name)** = example.com

Phishing URL Detection

10.5.1 Phishing URL Detection

- **Phishing** refers to attacks where a victim is lured to a fake web, and is deceived into disclosing personal data or credentials
 - Most common phishing scam tactics are shown below
- **Phishing URLs** seem like legitimate URLs, and redirect the users to phishing web pages, which mimic the look and feel of their target websites
 - E.g., a fake bank website hopes that the user will enter personal data (password)





Phishing Scam Statistics

10.5.1 Phishing URL Detection

- Google blocks around 100 million phishing emails every day
- 90% of phishing attacks sent via messaging apps are sent through WhatsApp
- Fake invoices are used in 26% of phishing scams
- Top 5 phishing targets in 2022 were:
 - LinkedIn – 52% (i.e., URLs with links that mimic the LinkedIn website)
 - DHL – 14%
 - Google – 7%
 - Microsoft – 6%
 - FedEx – 6%

Phishing URL Detection

10.5.1 Phishing URL Detection

- Phishing emails are a more serious threat than spam emails, because they aim to steal users' private information, such as bank accounts, passwords, SSNs
- Machine learning techniques are widely used to identify anomalous patterns in URLs as signs of possible phishing
 - Examples of such anomalous patterns are shown in the table
- ML-based phishing detection models are usually embedded in web browsers as extensions, or into email spam filtering systems
 - Thus, they appear as a black-boxes to phishers, as it is difficult to identify which features or classification algorithms they use

Clue in the URL	Example
Includes redirection	http://3104.nnu4urye.info?http://c43n34.com?35u3b
The path contains a URL of a known organization	http://108.179.216.140/~bankofamerica/
Special characters “-” in the host name	http://yj4yb6hmb3.x-cant-bank-you-here-of-my-money.cn/yj4yb6hmb3/Oraliao_show_23Y
Long domain name	http://31837.9hzaseruijintunhfeugandeikisn.com/5/54878
Hostname is Encoded	http://www.%64isc%72%65%74%2done-%6ei%67h%74.%63o%6d
IP is Encoded	http://0x42.0x1D.0x25.0xC2/
E-mail Address in URL	http://username@hotmail.com.fddcol.com

AML against Phishing URL Detectors

10.5.2 Adversarial Attacks Against Phishing URL Classifiers

- *Bahnsen et al. (2018) DeepPhish: Simulating Malicious AI* ([link](#))
 - White-box attack
 - Against a character-level LSTM-based phishing URL classifier
 - The classifier was trained using URLs from historical attacks
 - The LSTM model predicts the next character in the URL
 - The attack concatenates benign URLs to the phishing URLs to evade the classifier
 - The form of synthetic URLs is: `http:// + compromised_domain + benign_URL`
 - Limitation: concatenation of benign URLs can be signed, which makes the attack less effective for real URL detectors

AML against Phishing URL Detectors

10.5.2 Adversarial Attacks Against Phishing URL Classifiers

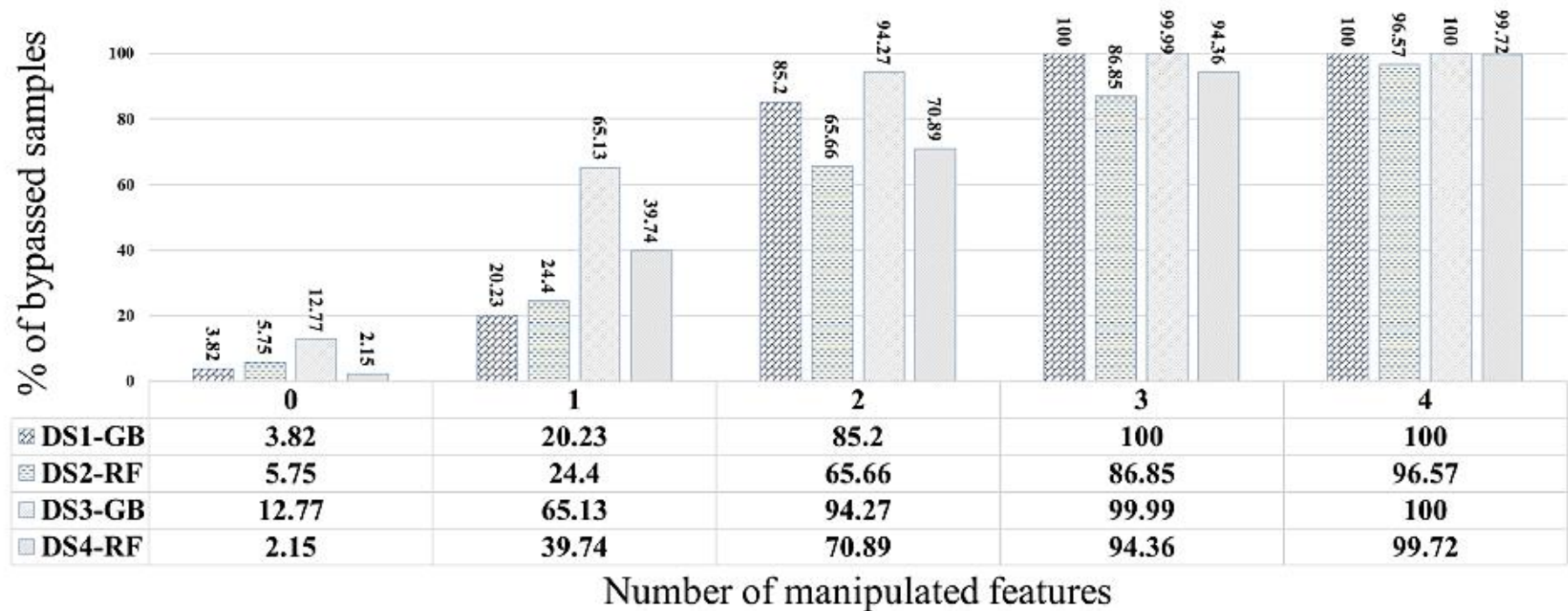
- *Shirazi et al. (2019) Adversarial Sampling Attacks Against Phishing Detection* ([link](#))
 - Gray-box attack, requires knowledge of the features used by the ML-based classifier
 - Such knowledge may not be accessible to the attacker, hence, this type of attacks may be less feasible in real-life scenarios
 - Eight features used: domain length, presence of non-alphabetic characters in domain name, ratio of hyperlinks referring to domain, presence of HTTPS protocol, matching domain name with copyright logo, and matching domain name with the page title
 - Try all possible combinations for the values of the features used by the classifier
 - An objective function minimizes: number of manipulated features + assigned feature values
 - The adversarial samples must be visually or functionally similar to the targeted websites
 - Characteristics of used datasets are shown in the table

Dataset	Data shape (#)		Instances (%)	
	Size	Features	Legitimate	Phishing
DS-1	2210	7	44.71	55.29
DS-2	11055	30	55.69	44.31
DS-3	1250	9	43.84	56.16
DS-4	10000	48	50.0	50.0

AML against Phishing URL Detectors

10.5.2 Adversarial Attacks Against Phishing URL Classifiers

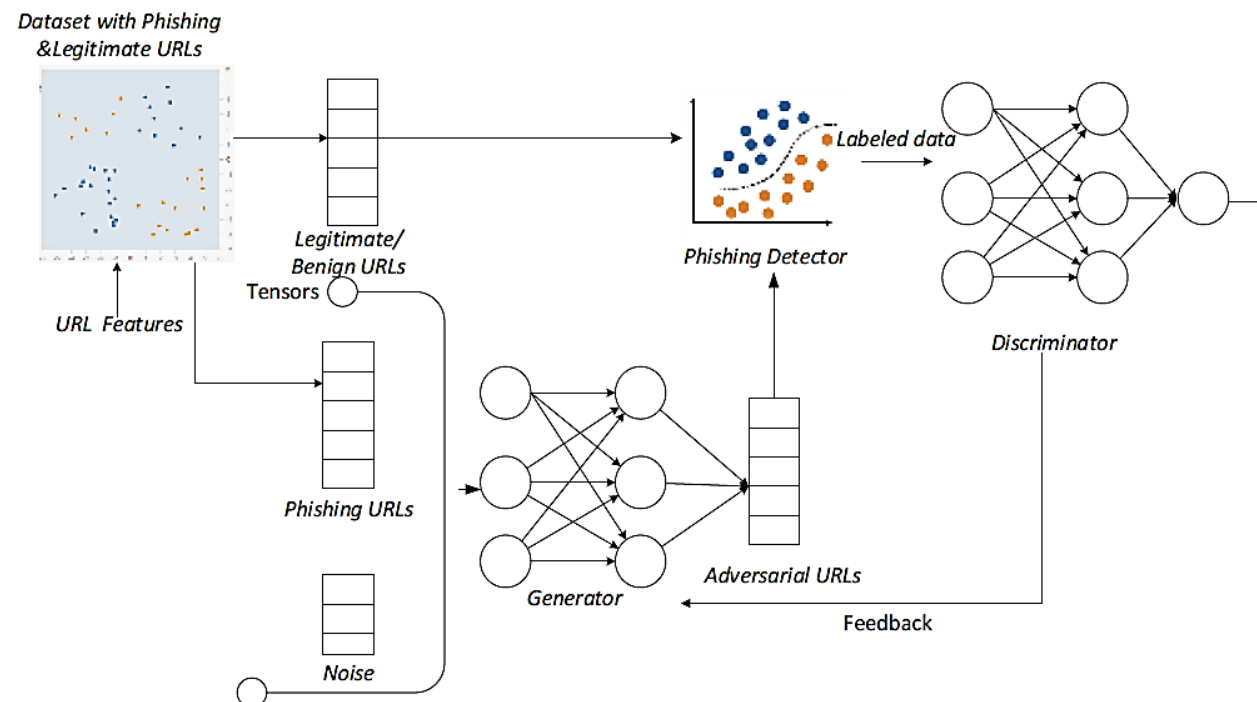
- Shirazi et al. (2019) cont'd
 - The percentage of evaded URL samples increased significantly with only one feature perturbed, and reached 100 when four features were manipulated



AML against Phishing URL Detectors

10.5.2 Adversarial Attacks Against Phishing URL Classifiers

- *AlEroud and Karabatis (2020) Bypassing Detection of URL-based Phishing Attacks Using Generative Adversarial Deep Neural Networks* ([link](#))
 - Black-box evasion attack
 - Employs a GAN model to generate phishing URLs to evade ML phishing detectors
 - The generator used a perturbed version of phishing URLs and converted them to adversarial examples



AML against Spam Filters

10.4.3 Adversarial Attacks Against ML-based Spam Filters

- *Apruzzese et al. (2022) SpacePhish: The Evasion-space of Adversarial Attacks against Phishing Website Detectors using Machine Learning* ([link](#))
 - Black-box evasion attack against ML-based phishing website detectors
 - The goal is to modify phishing URLs to evade detection while preserving functionality
 - Attack approach:
 - Feature-space attacks: directly manipulate extracted feature vectors ((not actual URLs) to cross the decision boundary
 - Example features include special characters, entropy, number of dots, etc.
 - Limitation is that it may produce unrealistic or non-functional URLs
 - Problem-space attacks: modify the content of URLs in a functionally valid way
 - E.g., add benign-looking tokens (e.g., “login”, “secure”), domain perturbations (typos, subdomains), pad URLs to mimic legitimate patterns
 - The main finding is that realistic problem-space attacks are highly effective and achieve high evasion rates against ML detectors
 - ML detectors are overly reliant on semantic word features, and are vulnerable to small, realistic perturbations

Cyber-physical Systems (CPS)

10.6 Cyber-physical Systems (CPS)

- *Cyber-physical systems (CPSs)* consist of hardware and software components that control and monitor physical processes
 - CPS are part of the *critical infrastructure*, which includes the electric power grid, transportation networks, water supply networks, nuclear plants, telecommunications, etc.
- The increased use of ML-based models for controlling and monitoring CPS makes these systems vulnerable to adversarial attacks
 - Adversarial manipulation of sensory data (if undetected) can cause substantial physical and financial damage
 - This can range from major disruptions, power blackouts, to nuclear incidents
 - For instance, AML attacks on manufacturing production systems can cause:
 - Damage to the systems, processes, and equipment
 - Defective products
 - Safety threat to employees
 - Lost production time
 - Unscheduled maintenance (due to false alarms)

Industrial Control Systems (ICS)

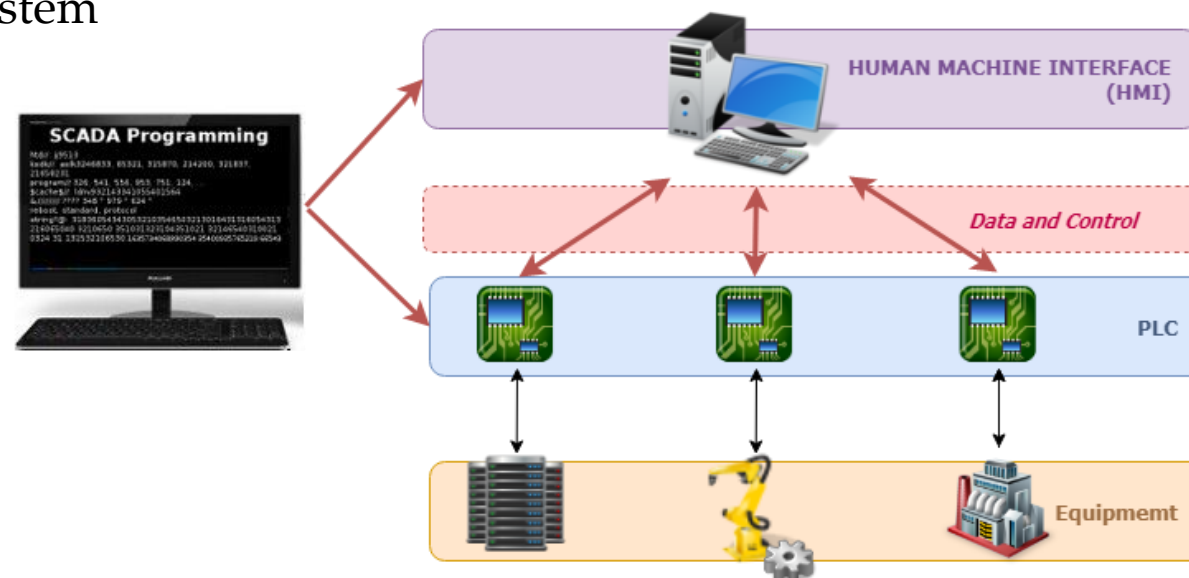
10.6 Cyber-physical Systems (CPS)

- **Industrial Control Systems (ICS)** are a subcategory of CPS
 - ICS commonly consist of sets of connected devices, such as PLCs, sensors, and actuators
- The recent trend of connected devices and cloud-based services in ICS exposes these systems to increased risk of cyber attacks
- Controllers make decisions for regulating process parameters based on readings from critical sensors
 - E.g., increase or decrease temperature to ensure it is within a target range
 - The ability of a malicious actor to manipulate sensory data in ICS can have catastrophic impacts on the control systems
- **Stealthy attacks** via injecting false sensory readings can cause the controller to place the system into an unsafe mode of operation
 - E.g., the attacker may compromise the computer network in a nuclear plant, and sends low temperature sensor readings, causing the controller to heat up the reactor above safe levels

Industrial Control Systems (ICS)

10.6 Cyber-physical Systems (CPS)

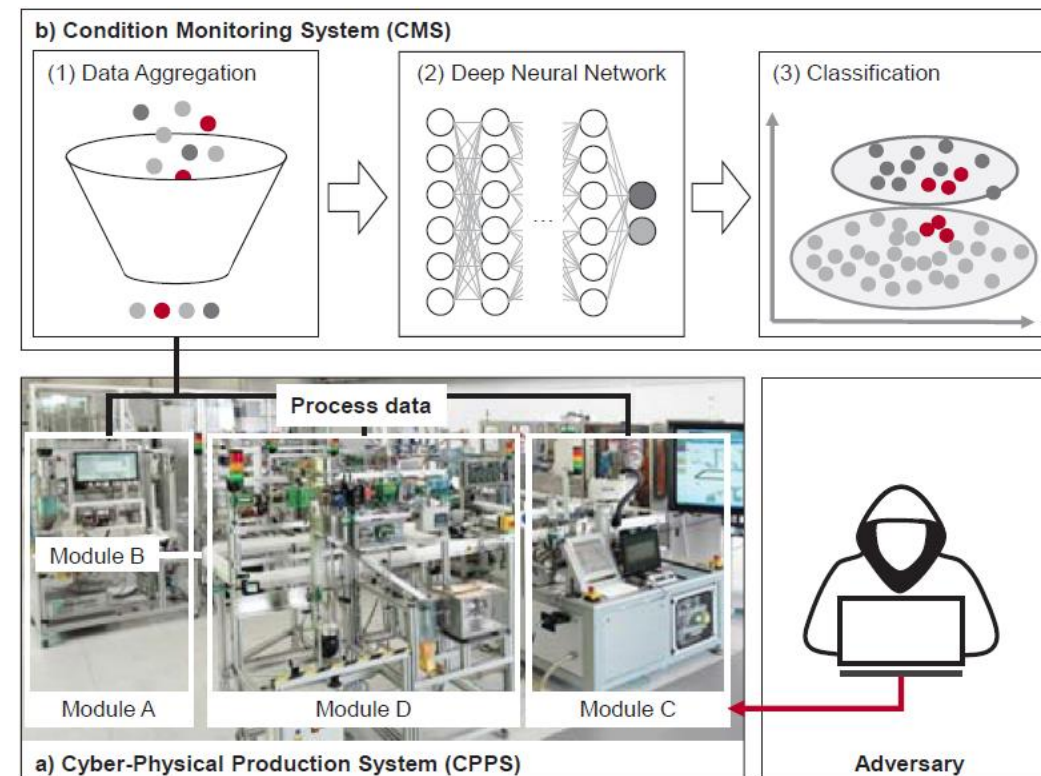
- The main components of ICS are depicted in the figure
 - Equipment** – includes various field devices have sensors and actuators (motors, hydraulic or pneumatic cylinders), such as robots, machines, CNCs, etc.
 - PLCs (Programmable Logic Controllers)** – industrial microcomputers that collect input data from local sensors and output control signals to actuators
 - SCADA (Supervisory Control And Data Acquisition)** – is a central computer station that gathers information from multiple PLCs and manages the operation of the system
 - HMI (Human Machine Interface)** – an interface for human operators to monitor and control the system



Machine Learning-based CPS

10.6.1 Machine Learning-based CPS

- *ML-based anomaly detection models* are commonly used for monitoring the conditions in CPS, and for detecting abnormal conditions or system failures
 - E.g., in the figure, the **condition monitoring system (CMS)** will stop the production process if one of the modules fails
- In this case, CMS is implemented by training the ML model on historical process data from production modules, to learn the characteristics of normal and abnormal system conditions
- Adversarial attacks can manipulate the physical system without being detected by the CMS
 - Another objective of the attacker may be to trigger false alarms (to temporary stop the production)



Major CPS Attacks

10.6 Cyber-physical Systems (CPS)

- List of the most notorious attacks on CPS
 - **Stuxnet (2010)** – a malware attack on ICS around the world, e.g., it destroyed uranium centrifuges in an Iranian nuclear plant via PLC manipulation
 - **New York dam attack (2013)** – a group of Iranian hackers accessed the Bowman Dam in New York, they didn't do any damage, but showed infrastructure vulnerability
 - **German steel mill (2014)** – the attackers caused extensive damage to the steel mill, by preventing the blast furnace from shutting down
 - **Ukraine power grid (2015 and 2016)** – a Russian-based cyber attack remotely disabled power stations and left about 250,000 customers without electricity for 6 hours (in 2015) and 3 hours (in 2016)
 - **Kemuri water plant (2016)** – attack on PLCs for controlling the valves used for water treatment chemical processing
 - **San Francisco municipality attack (2016)** – ransomware attack on the municipal transportation agency, the ticketing system was disrupted, about 1,000 computer were infected; didn't pay the ransom, used backups to restore system
 - **SolarWinds attack (2020)** – attack on IT management software, attackers inserted malware, updates distributed to 18,000 customers, the focus is on cyber espionage and not disruption

Major CPS Attacks

10.6 Cyber-physical Systems (CPS)

- **Water plant in Florida (2021)** – an attempted attack to poison the water by increasing the level of one chemical hundredfold, it was discovered immediately and corrected
- **JBS Meat Processor (2021)** – a ransomware attack on the world's largest meat processor forced shutdown on 9 plants in US, JBS paid the requested \$11M
- **Colonial Pipeline (2021)** – ransomware attack on the largest US fuel pipeline, caused 6 days shutdown and regional fuel shortages; the attackers cracked the password (no two-factor authentication); Colonial paid the requested \$4.4 M
- **Rural healthcare providers in USA (2016-present)** – ransomware attacks on many small healthcare providers with limited IT security and resources, usually computers locked, disrupted operations for weeks
- **Change Healthcare (2024)** – a ransomware attack on software for healthcare prescriptions and payment systems, the hackers were paid \$22 M
- **Airports cyberattack (2025)** – attack on several European airports via third-party IT provider, processing systems for passenger check-in and boarding were disrupted, hundreds of flights delayed or canceled; airports were not hacked directly but a shared service provider was compromised

AML against CPS

10.6.2 Adversarial Attacks Against CPS

- *Specht et al. (2018) Generation of Adversarial Examples to Prevent Misclassification of Deep Neural Network based Condition Monitoring Systems for Cyber-Physical Production Systems ([link](#))*
 - Application: monitoring a process for manufacturing semi-conductors
 - White-box evasion attack
 - Dataset: SECOM, recorded from a semi-conductor manufacturing process
 - Each data instance contains 590 features collected from the manufacturing sensors
 - The dataset contains 1,567 samples, labeled as either normal or anomalous production cycle
 - Attacked model: a deep NN consisting of fully-connected layers
 - The model is used for anomaly detection, i.e., it detects anomalous conditions in the collected sensory data
 - FGSM attack was used to generate adversarial samples, that were classified by the ML model as normal sensory data

AML against CPS

10.6.2 Adversarial Attacks Against CPS

- Specht et al. (2018) cont'd
 - The work also introduces a defense approach called **CyberProtect**
 - It uses the generated adversarial samples to retrain the DNN model (with both clean samples and adversarial samples)
 - This defense approach increased the classification accuracy of the DNN model from 20% to 82%

(1) Process data

Time	X	
09:59	0	
10:00	81.16	
10:01	42.33	
10:02	123.49	⋮
10:03	124.65	
10:04	50.81	
10:05	123.97	
10:06	0	
...		

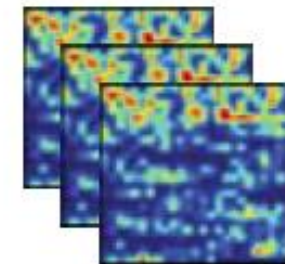


(2) CyberProtect

- (1) Select process data attributes
- (2) Select perturbation
- (3) Apply perturbation



(3) Adversarial Examples



Condition Monitoring



AML against CPS

10.6.2 Adversarial Attacks Against CPS

- *Ghafouri et al. (2018) Adversarial Regression for Detecting Attacks in Cyber-Physical Systems* ([link](#))
 - Application: controlling a process for manufacturing liquid products
 - Gray-box evasion attack
 - Attacked models: 3 ML models used for anomaly detection
 - These include: linear regression, NN, and an ensemble of LR and NN
 - Dataset: TE-PCS, containing sensory data from the production process
 - The data instances have 41 sensory measurements and 12 controlled outputs
 - Attack approach:
 - Mixed-integer linear programming (MILP) is used for generating adversarial examples
 - A challenge for this task is that there are safety constraints for the pressure and temperature readings of the sensors (the generated adversarial samples must obey these constraints)
 - Therefore, the attack was formulated as a constrained optimization problem via MILP

AML against CPS

10.6.2 Adversarial Attacks Against CPS

- *Feng et al. (2017) A Deep Learning-based Framework for Conducting Stealthy Attacks in Industrial Control Systems* ([link](#))
 - Applications: monitoring and controlling a small lab-scale gas pipeline, and a water treatment plant
 - Gray-box evasion attack
 - Attacked model: an LSTM-based anomaly detector
 - Datasets:
 - Gas pipeline data: 68,803 time series with 11 sensory measurement data
 - Water treatment plant dataset: 496,800 signals having 51 sensors measurements
 - Attack:
 - A GAN model is used to generate malicious sensor measurements to bypass the anomaly detector
 - Attacks on both sensor and control channels of the PLC were designed
 - The success rate can reach up to 90%, depending on the number of manipulated sensor measurements

AML against CPS

10.6.2 Adversarial Attacks Against CPS

- *Sheikh et al. (2023) Defending the Defender: Adversarial Learning-Based Defense Strategy for CPS Security* ([link](#))
 - Proposes defense strategy for CPS attacks based on adversarial training
 - Models evaluated: Random Forest, Neural Networks, LSTM
 - Attack approach
 - Generate adversarial data using GANs
 - Attack types: poisoning attacks with corrupt training data, and evasion attacks with small perturbations to bypass detection at inference
 - Defense strategy
 - Uses adversarial training framework by training models on both clean data and GAN-generated adversarial data
 - Findings:
 - ML models performance significantly degrades under adversarial attacks
 - GAN-generated adversarial data effectively simulates realistic attacks
 - Adversarial training improves robustness and results in more stable performance



Additional References

1. Rosenberg et al. (2021) – Adversarial Machine Learning Attacks and Defense Methods in the Cyber Security Domain ([link](#))
2. Ganagavarapu (2020) – Applicability of Machine Learning in Spam and Phishing Email Filtering: Review and Approaches ([link](#))
3. Blog Post by Vladislav Podolyako – Why Spam Filters Hate You ([link](#))
4. Daniel Lowd – Adversarial Machine Learning
5. Blog Post by Emily Bauer – 15 Outrageous Email Spam Statistics that Still Ring True in 2018 ([link](#))
6. Blog Post by Sie Huai Gan – How To Design A Spam Filtering System with Machine Learning Algorithm ([link](#))